E1-解放军总医院200P 小芯片Q30低

E1工单号： SO-2024020402-E1

武汉产品开发

现场案例分析总结

- 内容：

1. 现场问题描述

2. 客户端数据整理清单

3. 数据分析和排查方向

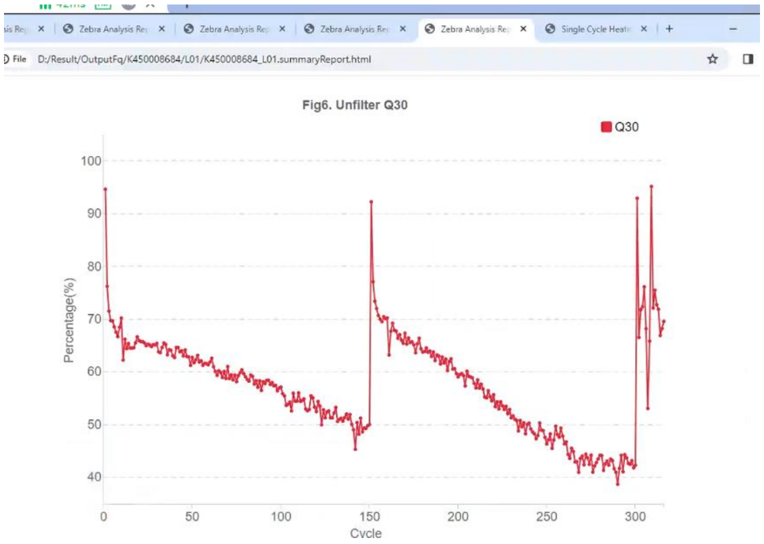
4. 现场FSE/FAS排查小结/现存关键问题梳理

5.下一步计划

01 现场问题描述

SN:20223003（艾德OEM） 200配置2 ECR5.0版本

客户跑FCS芯片，APP-E文库，PE150多次反馈有Q30偏低的情况，上机10次左右，5-6次都会出现



芯片号	日期	产出	Q30
S250059596	2023/7/14	608.7	93.68
K450005254	2023/8/11	145.18	85.86
K450008681	2023/10/21	21.6	78.38
K450008882	2023/10/28	127.16	85.04
K450008692	2023/11/3	145.19	86.97
K450008665	2023/11/11	53.49	77.15
K450008886	2023/11/18	137.87	97.71
K450010846	2023/12/9	40.5	78.83
K450008684	2023/12/16	62.92	80.5
S250070565	2023/12/24	654.6	流体 89.81
S250052836	2023/12/31	126.66	80.36
K450010931	2024/1/13	14.47	75.93

客户上机情况汇总，**频发数据差**

01 现场问题描述

华大智造 MGI

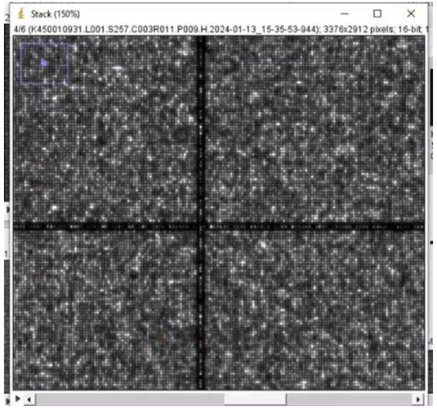
1.检查以往FCS数据异常下机报告表现一致，芯片利用率低，DNB信号差

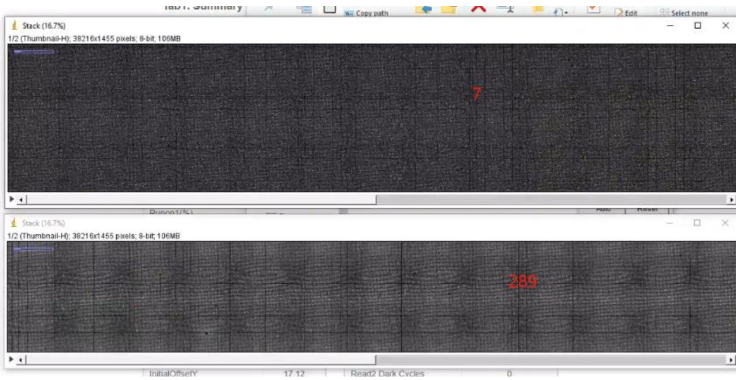
Tab1. Summary Information

Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	6.98
ImageArea	144
TotalReads(M)	14.47
Q30(%)	75.93
Runon1(%)	0.08
Runon2(%)	0.04
Lag1(%)	0.11
Lag2(%)	0.03
ESR(%)	6.98
[illegible]	33.90

Tab2. Biochemistry Information

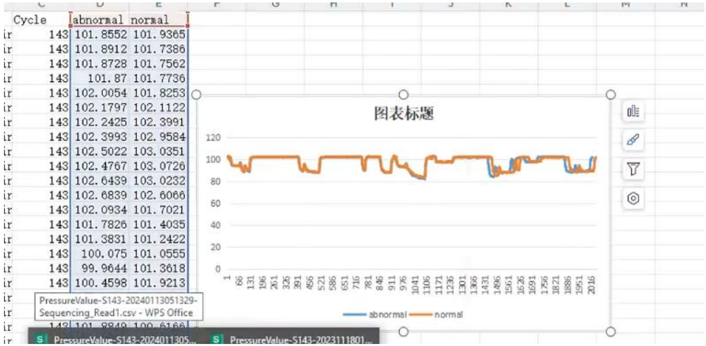
Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2024-01-12
Sequence Time	17:38:54
Reagent ID	CX2308080013
Flowcell Pos	A
DNB ID	20240112
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8





01 现场问题描述

2.对比正常run流体压力曲线无异常，查看缩略图热图都无流体痕迹



Cycle	abnormal	normal
143	101.8552	101.9365
143	101.8912	101.7386
143	101.8728	101.7562
143	101.87	101.7736
143	102.0054	101.8253
143	102.1797	102.1122
143	102.2425	102.3991
143	102.3993	102.9584
143	102.5022	103.0351
143	102.4767	103.0726
143	102.6439	103.0232
143	102.6839	102.6066
143	102.0934	101.7021
143	101.7826	101.4035
143	101.3831	101.2422
143	100.075	101.0555
143	99.9644	101.3618
143	100.4598	101.9213

3. 1月16日更换了出液口密封圈，定子转子，进出液块，bypass与bychip公共管路，注射器，16号试剂针，并校准校准激光功率和温控，问题仍存在

4.为验证仪器问题，跑标准文库PE100 FCL,下机数据正常,目前可以判断仪器问题概率低。

01 现场问题描述

5.已知艾德客户使用自配试剂

6.前端客户反馈信息如下：

1.客户端文库通过三方平行测试没有产出与质量问题。

2.客户端make DNB外送平行外送测试，没有产出问题。

3.app-E试剂在艾德内部全国范围内无风险批次问题。

4.客户端三轮FSE排查流体以及基本检查，但除16号孔有堵塞以外，修理后依然复现相同问题。

与艾德对接MGI负责人，以及艾德技术蒋老师沟通后，目前在推进客户端换设备工作。

下一步计划

目前售后初步判断客户文库与FCS芯片试剂存在不适配问题，需研发介入排查

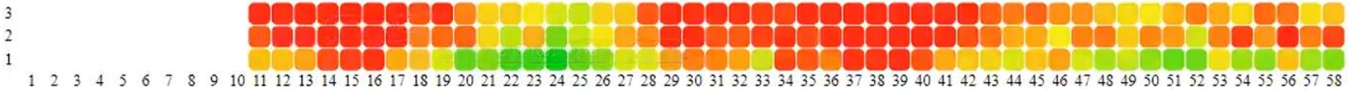
其他背景：
23年10月SN：20222001情况类似，测试标准文库没问题，但是测艾德的试剂就出现这种掉Q30的问题

艾德退机客诉

艾德PE150_FCS测试数据

K450008684

Figl. Cycle 23 HeatMap of Q30



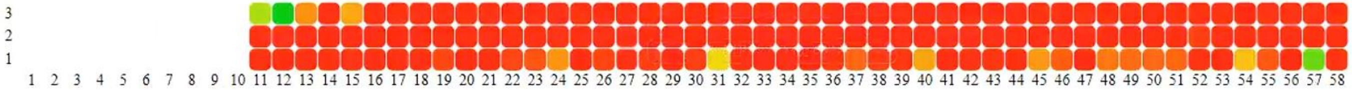
Figl. Cycle 252 HeatMap of Q30

数据量: 62.92 ESR：30.36

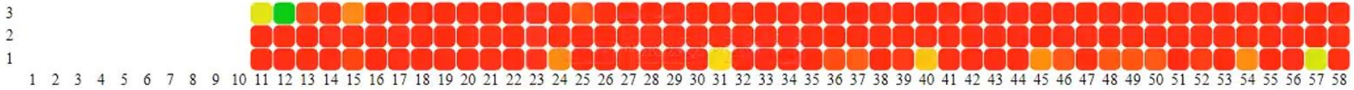
区域性-起始Q30偏低

K450010846

Figl. Cycle 25 HeatMap of Q30



Figl. Cycle 162 HeatMap of Q30



数据量:40.50 ESR：19.54

整体起始Q30偏低

K450008665

Figl. Cycle 3 HeatMap of Q30

Figl. Cycle 130 HeatMap of Q30

数据量: 53.49 ESR： 25.81

起始Q30正常 后续cycle下降快

艾德正常PE150_FCS测试数据

Tab1. Summary Information

Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	70.06
ImageArea	144
TotalReads(M)	145.19
Q30(%)	86.97
Runon1(%)	0.06
Runon2(%)	0.07
Lag1(%)	0.08
Lag2(%)	0.10
ESR(%)	70.06
MaxOffsetX	36.40
MaxOffsetY	21.97
InitialOffsetX	23.14
InitialOffsetY	14.33
RecoverValue(A_H)	1.37
RecoverValue(A_L)	0.98
RecoverValue(C_H)	1.31
RecoverValue(T_L)	0.96
RecoverValue(AVG)	1.16

Tab2. Biochemistry Information

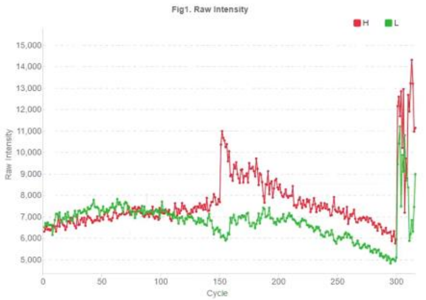
Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2023-11-02
Sequence Time	17:45:25
Reagent ID	CX2308080020
Flowcell Pos	A
DNB ID	20231102
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

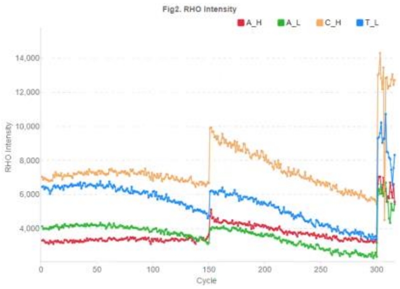
K4500008692

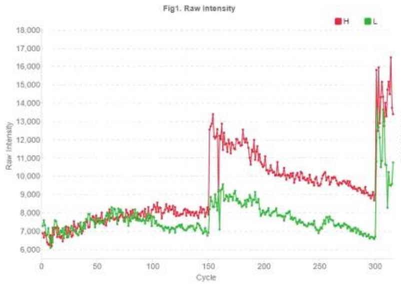
Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	66.53
ImageArea	144
TotalReads(M)	137.87
Q30(%)	87.71
Runon1(%)	0.06
Runon2(%)	0.07
Lag1(%)	0.08
Lag2(%)	0.10
ESR(%)	66.53
MaxOffsetX	37.08
MaxOffsetY	21.67
InitialOffsetX	25.41
InitialOffsetY	15.49
RecoverValue(A_H)	1.72
RecoverValue(A_L)	1.27
RecoverValue(C_H)	1.70
RecoverValue(T_L)	1.24
RecoverValue(AVG)	1.48

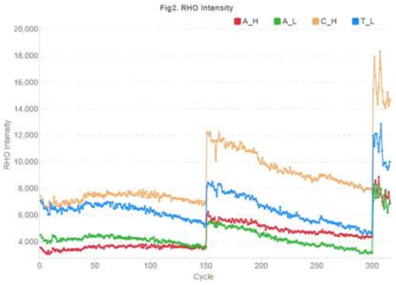
Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2023-11-17
Sequence Time	14:17:06
Reagent ID	CX2308080001
Flowcell Pos	A
DNB ID	20231117
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

K4500088886







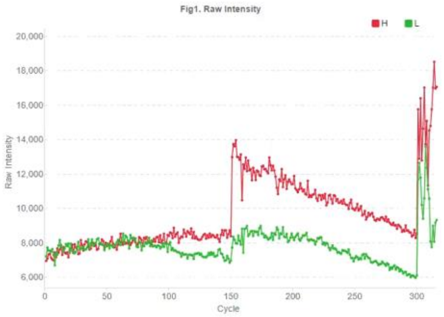


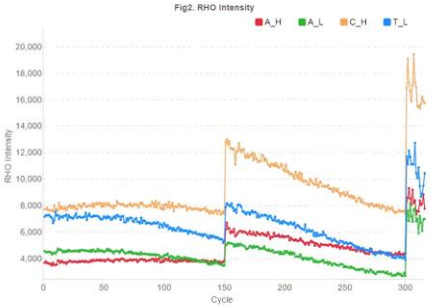
艾德正常PE150_FCS测试数据

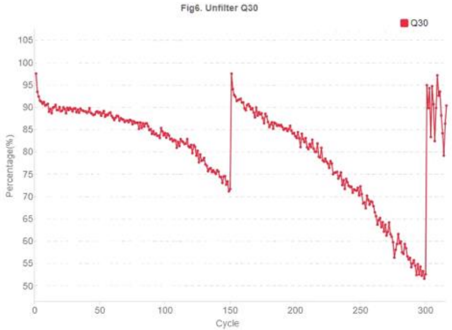
Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	71.83
ImageArea	144
TotalReads(M)	148.86
Q30(%)	89.57
Runon1(%)	0.06
Runon2(%)	0.07
Lag1(%)	0.09
Lag2(%)	0.11
ESR(%)	71.83
MaxOffsetX	36.61
MaxOffsetY	30.16
InitialOffsetX	25.63
InitialOffsetY	14.88
RecoverValue(A_H)	1.63
RecoverValue(A_L)	1.10
RecoverValue(C_H)	1.56
RecoverValue(T_L)	1.08
RecoverValue(AVG)	1.34

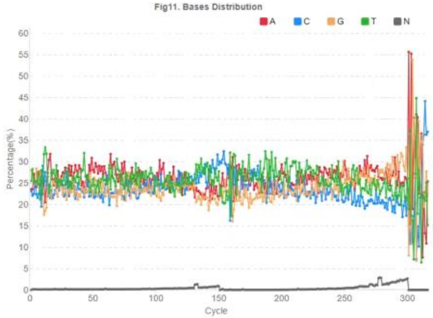
Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2024-02-02
Sequence Time	14:26:09
Reagent ID	CX2310310007
Flowcell Pos	A
DNB ID	20240202
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

K450010977









艾德异常PE150_FCS测试数据--二链异常

Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	25.81
ImageArea	144
TotalReads(M)	53.49
Q30(%)	77.15
Runon1(%)	0.07
Runon2(%)	0.05
Lag1(%)	0.10
Lag2(%)	0.05
ESR(%)	25.61
MaxOffsetX	34.05
MaxOffsetY	22.61
InitialOffsetX	21.92
InitialOffsetY	18.08
RecoverValue(A_H)	0.74
RecoverValue(A_L)	0.57
RecoverValue(C_H)	0.70
RecoverValue(T_L)	0.55
RecoverValue(AVG)	0.64

Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2023-11-10
Sequence Time	12:54:18
Reagent ID	CX2308080009
Flowcell Pos	A
DNB ID	20231110
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

K450008665

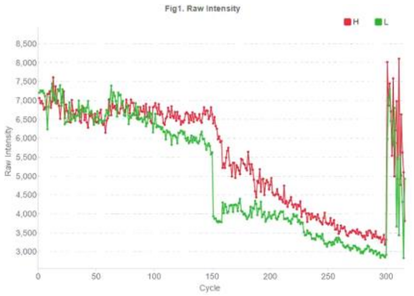
Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	6.98
ImageArea	144
TotalReads(M)	14.47
Q30(%)	75.93
Runon1(%)	0.08
Runon2(%)	0.04
Lag1(%)	0.11
Lag2(%)	0.03
ESR(%)	6.98
MaxOffsetX	33.90
MaxOffsetY	20.14
InitialOffsetX	24.09
InitialOffsetY	17.12
RecoverValue(A_H)	0.73
RecoverValue(A_L)	0.54
RecoverValue(C_H)	0.67
RecoverValue(T_L)	0.51
RecoverValue(AVG)	0.61

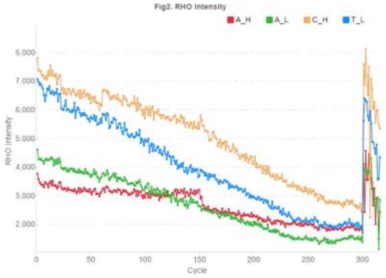
Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2024-01-12
Sequence Time	17:38:54
Reagent ID	CX2308080013
Flowcell Pos	A
DNB ID	20240112
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

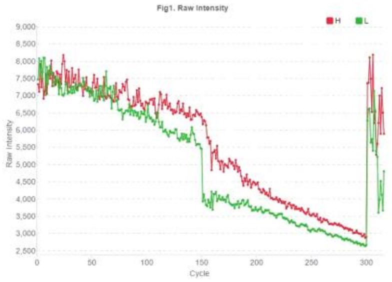
K4500109031

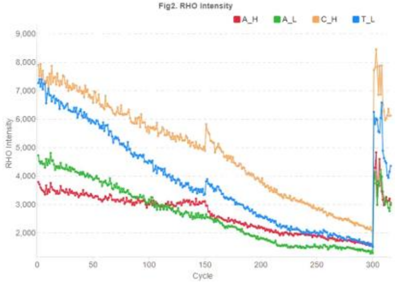
二链信号回升比偏低

PE合成异常









艾德异常PE150_FCS测试数据-区域FOV起始Q30偏低

Category	Value
SoftwareVersion	1.5.0.296
TemplateVersion	0.8.0
Reference	NULL
CycleNumber	316
ChipProductivity(%)	30.36
ImageArea	144
TotalReads(M)	62.92
Q30(%)	80.50
Runon1(%)	0.06
Runon2(%)	0.06
Lag1(%)	0.09
Lag2(%)	0.09
ESR(%)	30.36
MaxOffsetX	33.76
MaxOffsetY	25.07
InitialOffsetX	20.13
InitialOffsetY	16.54
RecoverValue(A_H)	1.90
RecoverValue(A_L)	1.32
RecoverValue(C_H)	1.81
RecoverValue(T_L)	1.32
RecoverValue(AVG)	1.59

Category	Value
ISW Version	1.1.0.1650
Machine ID	V20223003
Sequence Type	Customize
Recipe Version	
Sequence Date	2023-12-15
Sequence Time	15:56:33
Reagent ID	CX2310310032
Flowcell Pos	A
DNB ID	20231215
Barcode Type	others
Barcode File	
Read1 Cycles	150
Read2 Cycles	150
Barcode	8
Dual Barcode	8
Read1 Dark Cycles	0
Read2 Dark Cycles	0

K450008684

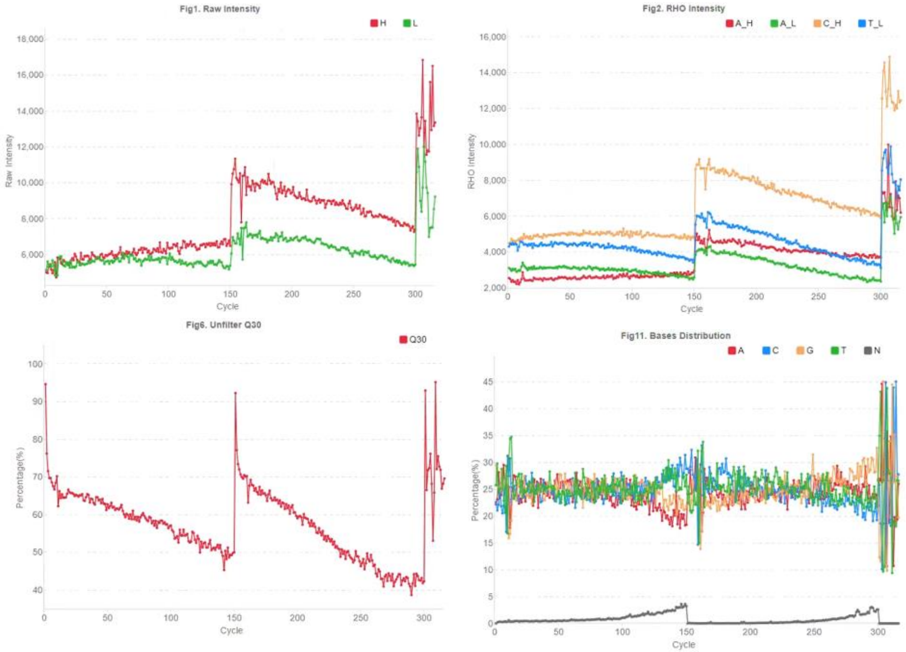
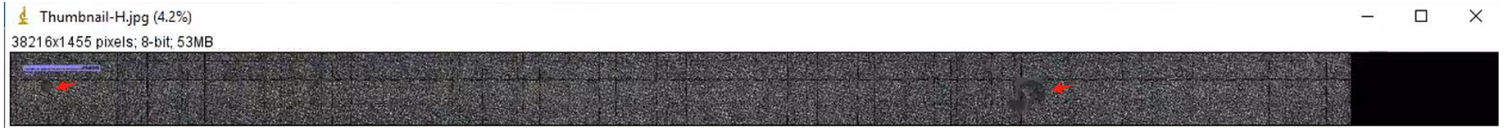
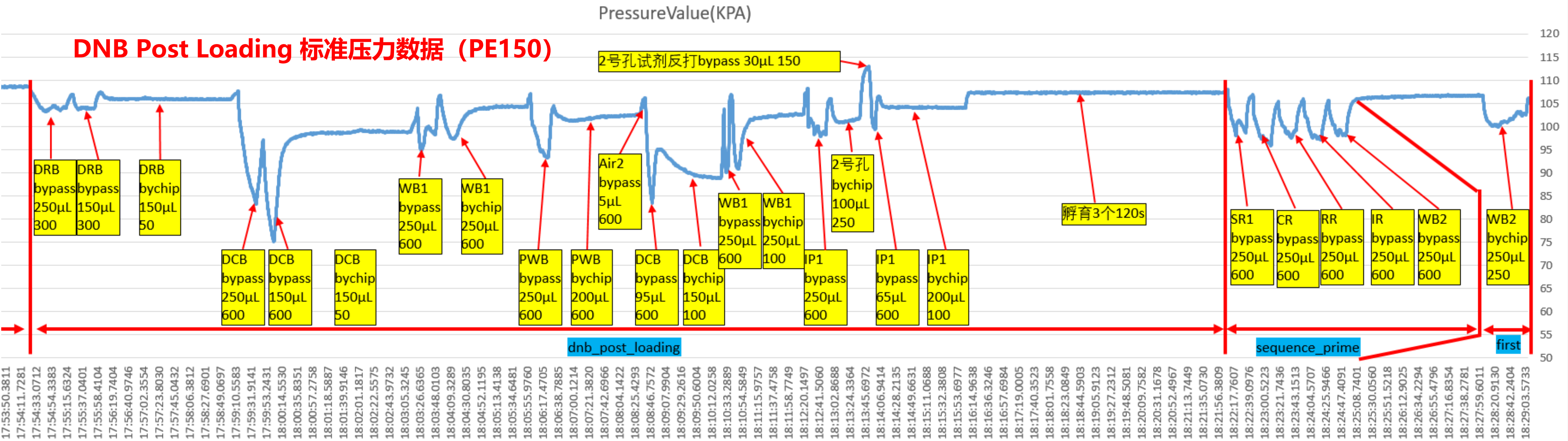


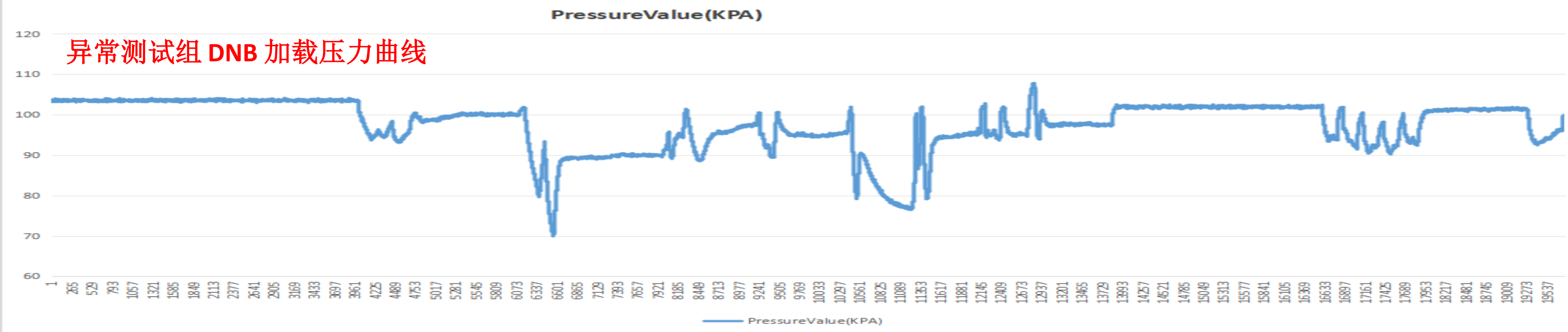
Fig1. Cycle 2 HeatMap of Q30



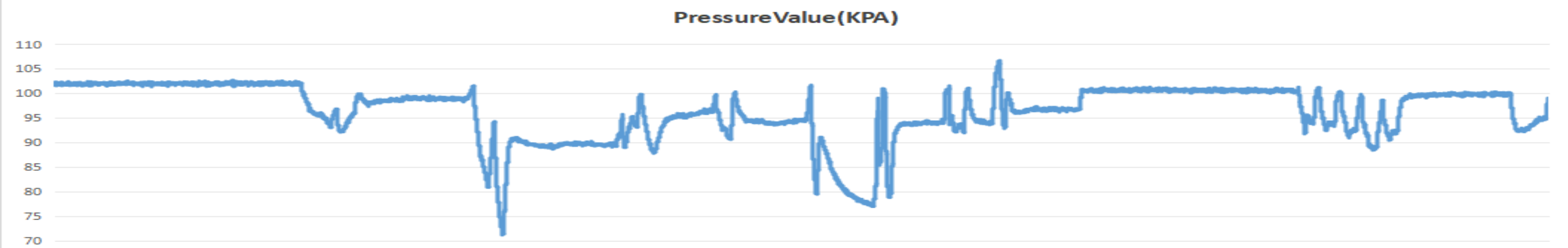
区域FOV
起始Q30偏低

G50流体压力流体压力曲线参考数据

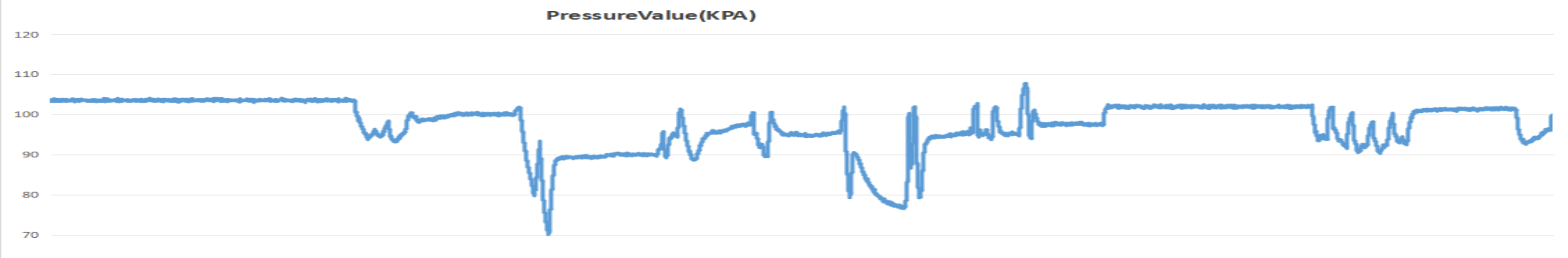




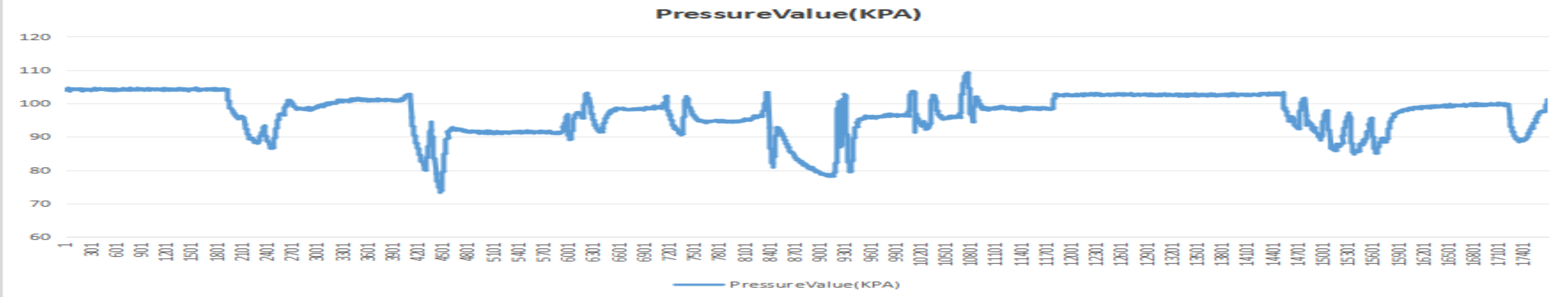
正常VS异常测试组 DNB加载压力曲线--无明显差异



正常

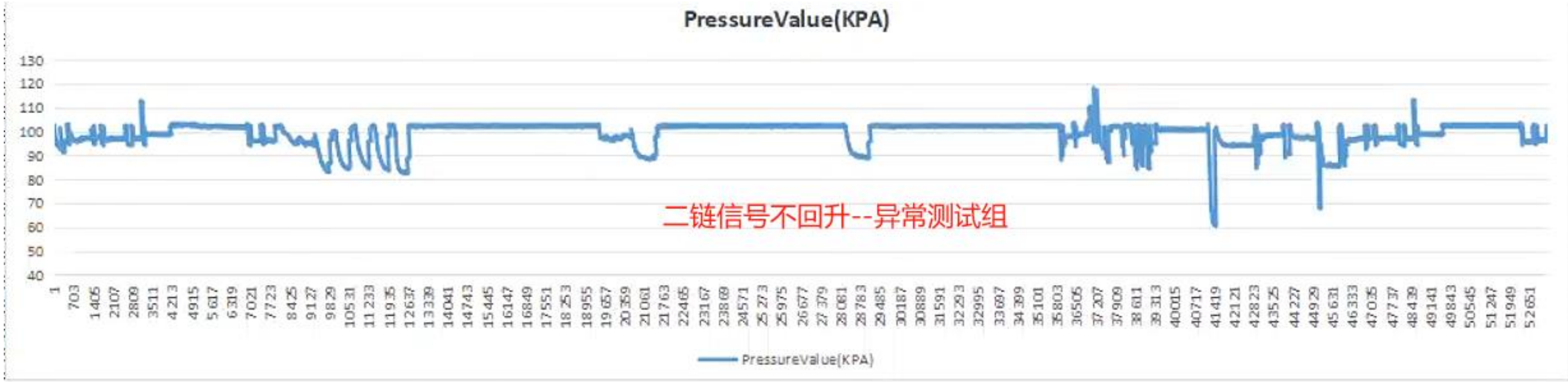


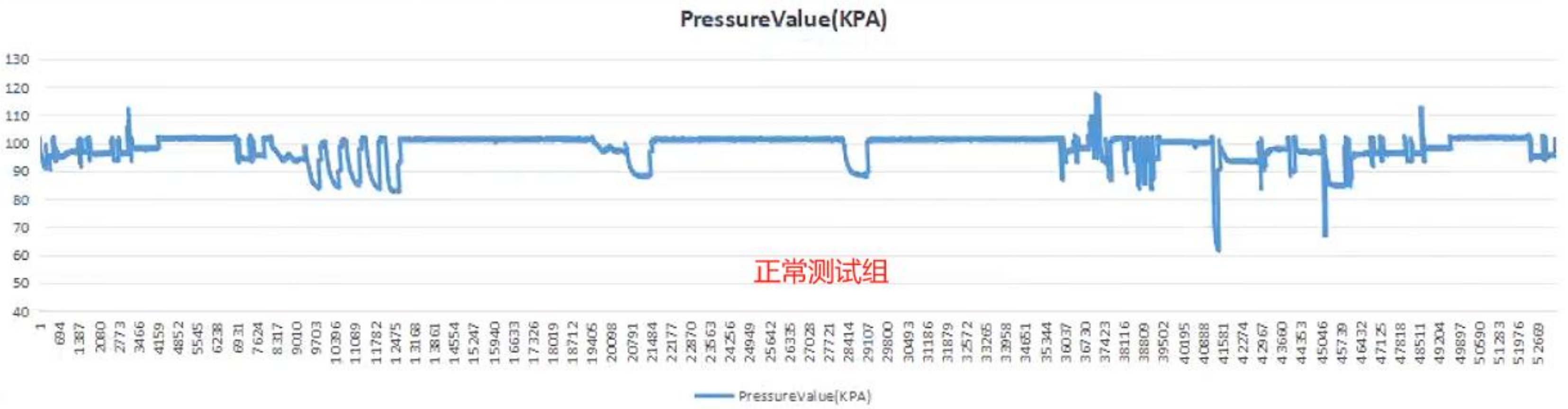
异常



异常

正常VS异常测试组 PE合成压力曲线--无明显差异





小结

- **K45008665/K450010931**异常测试均表现**二链信号不回升**，对比正常PE合成压力曲线无明显差异，推测其对应的试剂耗材出现异常概率更高；
- **K450008684**异常测试二链信号正常回升，但热图表现**区域FOV加载异常**，缩略图局部Q30偏低；
- **K450010846**异常测试二链信号回升正常，但热图表现**前几cycle整体FOV Q30偏低**，推测文库质量/DNB制备异常;

Thanks for your time!

感谢！

华大智造

MGI

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